O-RAN.WG10.TS.O1NRM.0-R005-v05.00

Technical Specification

O-RAN Work Group 10 (OAM for O-RAN)

O1 Network Resource Model

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Contents

Foreword

This Technical Specification (TS) has been produced by WG10 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the Information Model and the Data Model for Network Resource Model (NRM) that are foundational for functions carried out over the O-RAN O1 interface.

The O-RAN Information Model follows the methodology documented in 3GPP TS 32.160 [1] Clause 5.2.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in 3GPP Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] 3GPP TS 32.160: “Management and orchestration; Management service template”

[2] 3GPP TS 32.156: “Telecommunication management; Fixed Mobile Convergence (FMC); Model repertoire”

[3] 3GPP TS 28.622: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)", v18.8.0

[4] 3GPP TS 28.625: "Telecommunication management; State Management data definition Integration Reference Point (IRP); Information Service (IS)"

[5] O-RAN.WG3.TS.E2AP: "Near-Real-time RAN Intelligent Controller and E2 Interface -E2 Application Protocol (E2AP)"

[6] O-RAN.WG3.TS.RICARCH: " Near-RT RIC Architecture"

[7] 3GPP TS 28.658: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)"

[8] ITU-T Recommendation X.731: "Information technology - Open Systems Interconnection - Systems Management: State management function"

[9] O-RAN.WG1.TS.OAD: "O-RAN Architecture Description"

[10] 3GPP TS 28.532: “Management and orchestration; Generic management services”

[11] O-RAN.WG10.TS.O1-Interface: “O-RAN O1 Interface Specification”

[12] 3GPP TS 28.541: “Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3”

[13] O-RAN.WG4.TS.CUS.0: “Control, User and Synchronization Plane Specification”

[14] 3GPP TS 38.211: “NR; Physical channels and modulation”

[15] Semantic Versioning Specification 2.0.0: https://semver.org

[16] 3GPP TS 28.111: “Management and orchestration; Fault Management (FM)”

[17] O-RAN.WG10.TS.Information Model and Data Models: “O-RAN Information Model and Data Model”

[18] O-RAN.WG5.TS.D2GAP: “D2 Interface: General Aspects and Principles”

[19] O-RAN.WG5.TS-D2-O&MRequirements: “D2 Interface: O&M Requirements”

[20] O-RAN.WG5.TS.D2AP: “D2 Application Protocol (D2AP)”

[21] 3GPP TS 38.300: "NR; Overall description; Stage-2"

[22] IETF RFC 791: "Internet Protocol"

[23] IETF RFC 2373: “IP Version 6 Addressing Architecture”

[24] O-RAN.WG4.TS.MP.0: “Management Plane Specification”

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in 3GPP Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] 3GPP TR 21.905: “Vocabulary for 3GPP Specifications”

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in [i.1] and the following apply:

Information Model: a representation of concepts and the relationships, constraints, rules, and operations to specify data semantics for a chosen domain of discourse, it specifies relations between objects, can provide sharable, stable, and organized structure of information requirements or knowledge for the domain context.

Data Model: an abstract model that organizes elements of data and standardizes how they relate to one another and to the properties of real-world entities. The term data model may refer to two distinct but closely related concepts: (1) an abstract formalization of the objects and relationships found in a particular application domain; (2) the set of concepts used in defining such formalizations - for example concepts such as entities, attributes, relations, or tables.

D2 interface: The intra O-CU inter O-DU interface to enable multi-vendor carrier aggregation as defined in O-RAN Architecture Description [9], clause 5.4.22.

PCell: Primary cell

SCell: Secondary cell

PCell O-DU: O-DU, where the PCell is configured for a UE to support inter O-DU carrier aggregation.

SCell O-DU: O-DU, where only SCell(s) are configured for a UE to support inter O-DU carrier aggregation.

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in [i.1] and the following apply:

BS	Base Station
CRB	Common Resource Block
C-RNTI	Cell Radio Network Temporary Identifier
D2-c	D2 Control Plane interface
D2CP	D2 Control Plane Protocol
D2-u	D2 User Plane interface
D2UP	D2 User Plane Protocol
DL	Down Link
DN	Distinguished Name
HARQ	Hybrid Automatic Repeat Request
IOC	Information Object Class
MnS	Management Service
MOI	Managed Object Instance
NES	Network Energy Saving
NR	New Radio
NRM	Network Resource Model
O-DU	O-RAN Distributed Unit
O-RAN	Open Radio Access Network
O-RU	O-RAN Radio Unit

4 Requirements

4.1 General Requirements

The following general requirements apply:

Table 4.1-1: General Requirements

Requirement label	Description	Motivation
REQ-NRM-MC-1	O1 NRM stage 2 shall follow the templates described in clause 5.2 of 3GPP TS 32.160 [1], using the applicable fonts defined in Table 5.1.1-1 of 3GPP TS 32.160 [1].	Identify the templates and fonts to be used for the definition of O1 network resource model
REQ-NRM-MC-2	The O1 NRM stage 2 shall follow the model repertoire documented in 3GPP TS 32.156 [2]	Identify common set of UML notations to model network resources
REQ-NRM-MC-3	The mapping rules (traceability) defined in O-RAN WG10 IM/DM TS [17] clause 6.2, shall be applicable to O-RAN O1 YANG modules. In case of ambiguity, the definition in this document takes precedence.	Identify traceability rules for O1 YANG modules
REQ-NRM-MC-4	In addition to the naming rules defined in O-RAN WG10 IM/DM TS [17] clause 6.2.2, following extensions shall be applied to O1 YANG modules: • MODULE NAMING: The <identifier> element is mandatory and shall have the value of “o1”. • PREFIX NAMING: The <org> element is mandatory and shall have the value of “or”. The <abbreviatedidentifier> element is mandatory and shall have the value of “o1”.	Identify naming rules for O1 YANG modules
REQ-NRM-MC-5	The rules defined in 3GPP TS 32.160 [1] clause 6.2 shall be applicable to O-RAN YANG modules except for the following ones: clause 6.2.1.1 Modeling Resources, 6.2.1.2 Unique YANG Module names, 6.2.1.3 Unique YANG Namespace, 6.2.1.4 Unique YANG Module Prefixes, 6.2.1.11 Module header statements, 6.2.1.12 Provide description and reference statements, and 6.2.1.19 Copyright. In case of ambiguity, the definition in this document takes precedence, extended by the ones in O-RAN WG10 IM/DM TS [17] clause 6.2, as in REQ-NRM-MC-3.	Identify the rules for the development of YANG data models compatible with O1
REQ-NRM-MC-6	In addition to the rules defined in O-RAN WG10 IM/DM TS [17] clause 6.2.8, the following applies: All O-RAN augmentations of 3GPP YANG Data Models shall follow the rules mentioned in REQ-NRM-MC-5. Vendor specific augmentation to O-RAN YANG modules shall be made in separate vendor originated YANG Data Model files and, shall follow 3GPP TS 32.160 [1], clause 6.2.1.8.	Identify augmentation of O1 YANG data modules

5 Model

5.1 Imported and associated information entities

5.1.1 Imported information entities and local labels

Table 5.1.1-1: Imported information entities and local labels

Label reference	Local label
3GPP TS 28.622 [3], IOC, *Top*	*Top*
3GPP TS 28.625 [4], Archetype, StateManagementEntity	StateManagementEntity
3GPP TS 28.622 [3], IOC, *ManagedFunction*	*ManagedFunction*
3GPP TS 28.622 [3], IOC, *EP_RP*	*EP_RP*
3GPP TS 28.658 [7], dataType, PLMNId	PLMNId
3GPP TS 28.622 [3], dataType, TimeWindow	TimeWindow
3GPP TS 28.541 [12], dataType, RRMPolicyMember	RRMPolicyMember
3GPP TS 28.541 [12], dataType, AddressWithVlan	AddressWithVlan

5.1.2 Associated information entities and local labels

Table 5.1.2-1: Associated information entities and local labels

Label reference	Local label
3GPP TS 28.622 [3], IOC, ManagedElement	ManagedElement
3GPP TS 28.541 [12], IOC, GNBDUFunction	GNBDUFunction
3GPP TS 28.541 [12], IOC, NRCellDU	NRCellDU
3GPP TS 28.541 [12], IOC, ExternalGNBDUFunction	ExternalGNBDUFunction

5.2 Class diagrams

5.2.1 Relationships

This clause depicts the set of classes (e.g. IOCs) that encapsulates the information relevant for this MnS. This clause provides an overview of the relationships between relevant classes in UML. Subsequent clauses provide more detailed specification of various aspects of these classes.

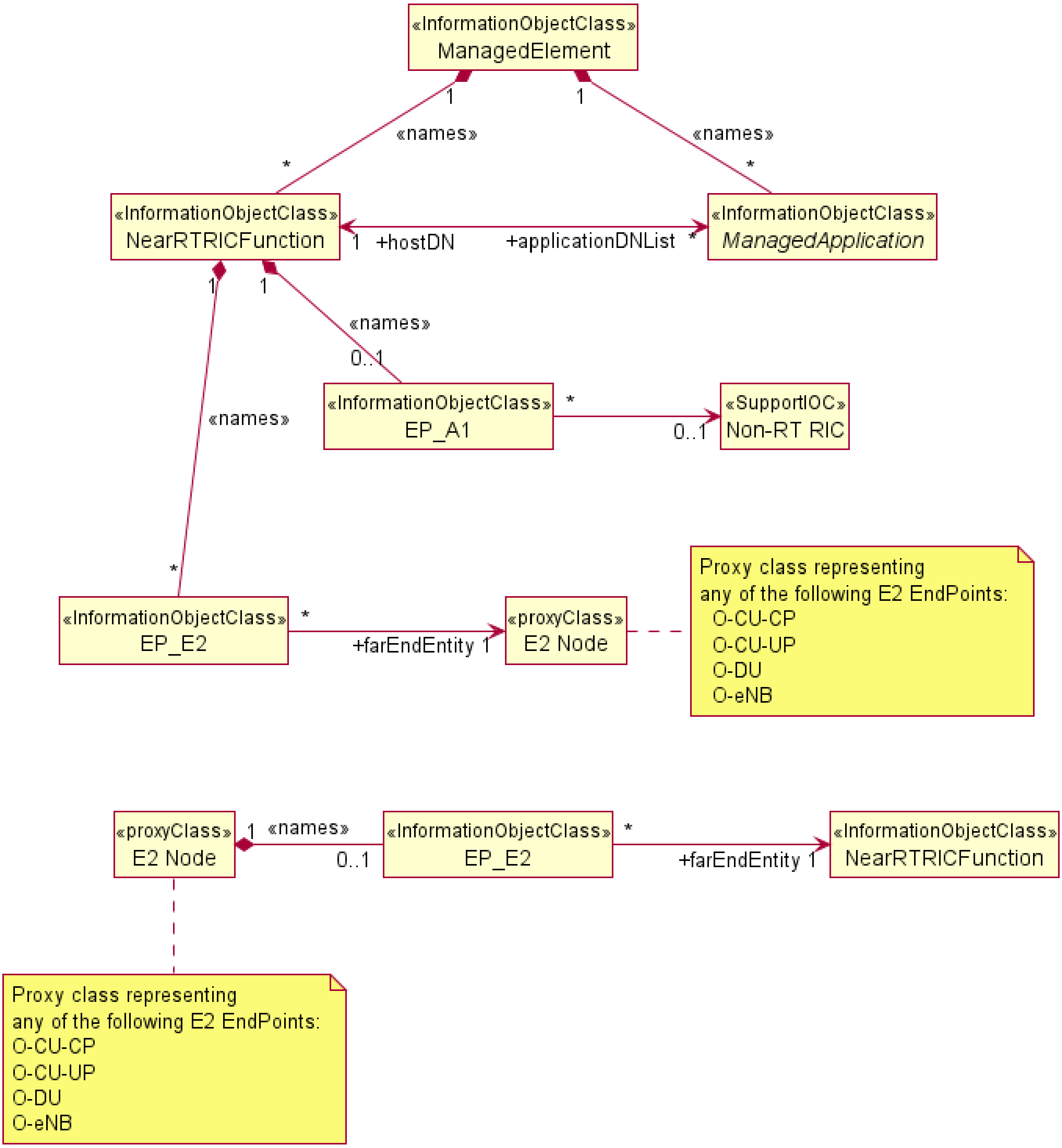

Figure 5.2.1-1: NRM for ManagedApplication and NearRTRICFunction related entities

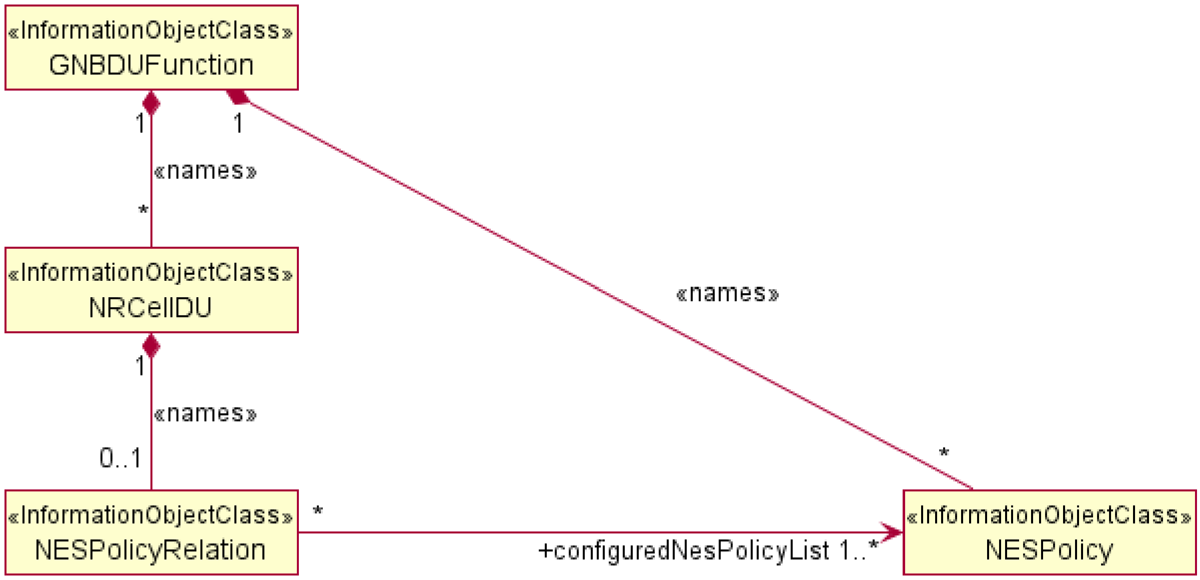

Figure 5.2.1-2: Network energy saving NRM fragment.

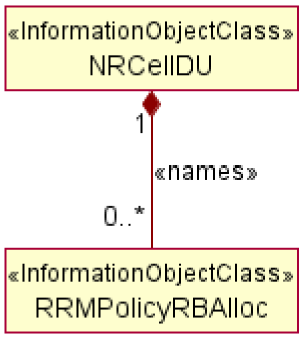

Figure 5.2.1-3: RRM Policy RB Allocation containment

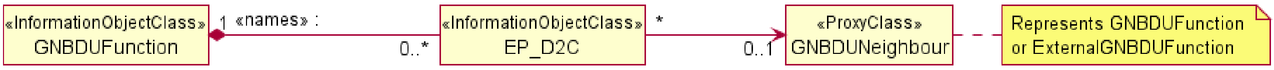

Figure 5.2.1-4: NRM for D2_C interface end points

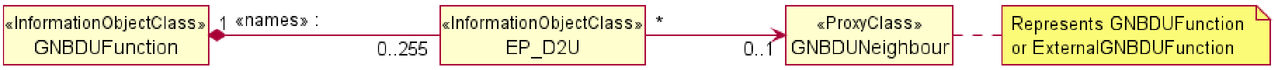

Figure 5.2.1-5: NRM for D2_U interface end points

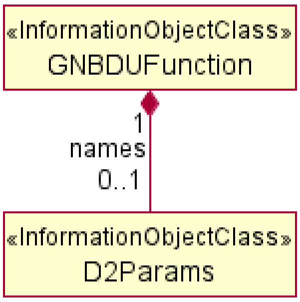

Figure 5.2.1-6: O-DU D2 parameters related containment

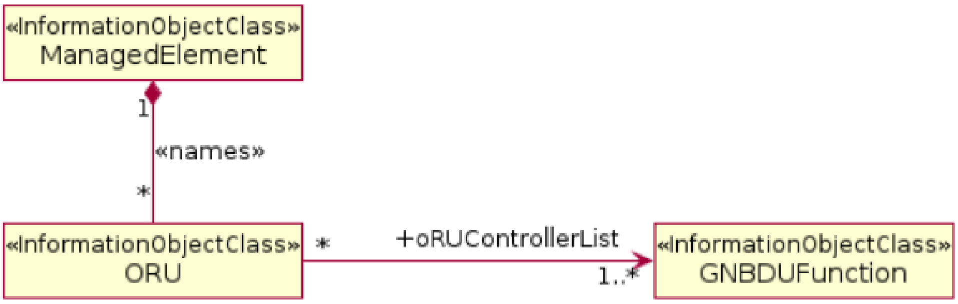

Figure 5.2.1-7: ORU containment and relationship

5.2.2 Inheritance

This subclause depicts the inheritance relationships.

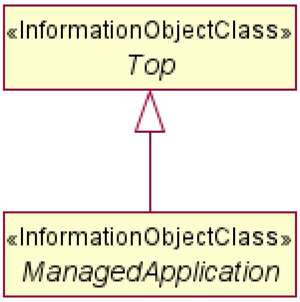

Figure 5.2.2-1: ManagedApplication inheritance

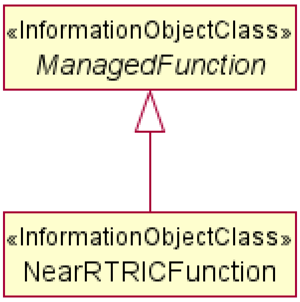

Figure 5.2.2-2: NearRTRICFunction inheritance

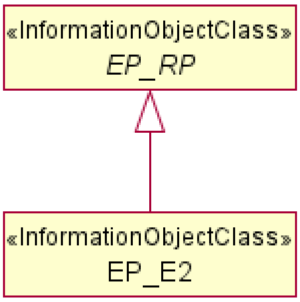

Figure 5.2.2-3: EP_E2 inheritance

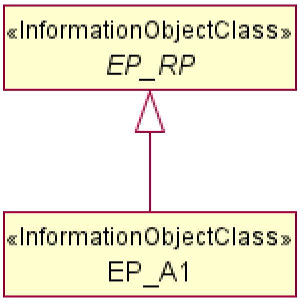

Figure 5.2.2-4: EP_A1 inheritance

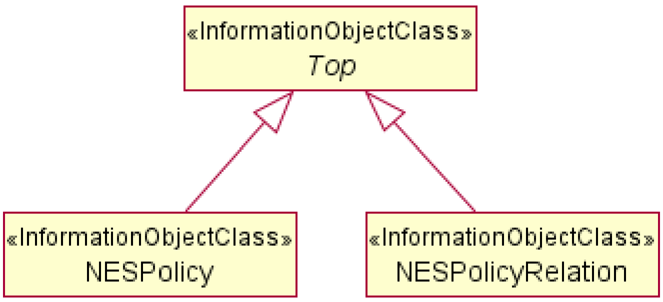

Figure 5.2.2-5: Network energy saving NRM inheritance.

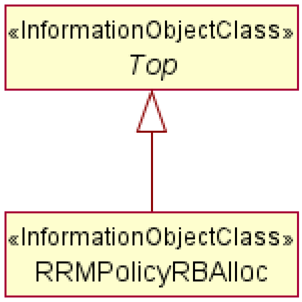

Figure 5.2.2-6: RRM Policy RB Allocation inheritance

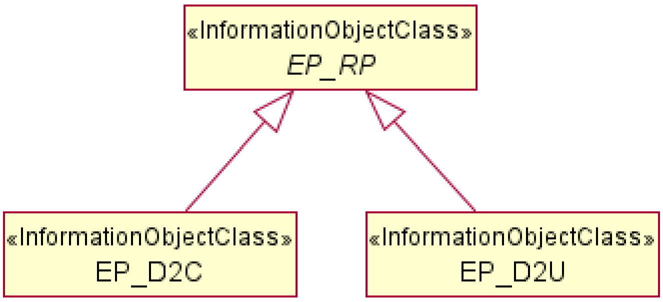

Figure 5.2.2-7: EP_D2C and EP_D2U inheritance

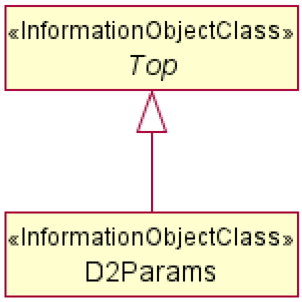

Figure 5.2.2-8: D2Params Inheritance

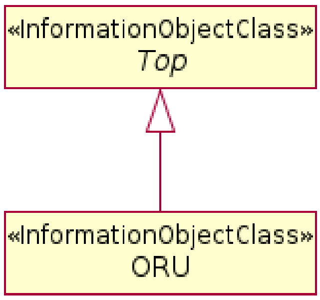

Figure 5.2.2-9: ORU inheritance

5.3 Class definitions

5.3.1 ManagedApplication

5.3.1.1 Definition

The *ManagedApplication* IOC defines attribute(s) that are common to application IOCs. This IOC represents a deployed instance of software application that may be independently tested and separately deployed from the hosting entity.

5.3.1.2 Attributes

The *ManagedApplication* IOC includes the inherited from *Top* IOC (defined in 3GPP TS 28.622 [3], clause 4.3.29), attributes operationalState, usageState, administrativeState imported from StateManagementEntity Archetype (defined in 3GPP TS 28.625 [4], clause 4.3.1) and the following attributes:

Table 5.3.1.2-1: ManagedApplication attributes

Attribute name	S	isReadable	isWritable	isInvariant	isNotifyable
applicationVersion	M	T	F	F	T
applicationName	M	T	F	T	T
userLabel	O	T	T	F	T
operationalState	M	T	F	F	T
usageState	M	T	F	F	T
administrativeState	M	T	T	F	T
Attribute related to role					
hostDN	M	T	F	T	T

5.3.1.3 Attribute constraints

None

5.3.1.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.1.5 State diagram

None

5.3.2 NearRTRICFunction

5.3.2.1 Definition

The NearRTRICFunction IOC represents the Management aspects of the aggregated functions making up the Near-RT RIC (defined in O-RAN Architecture Description [9], clause 5.3.2).

5.3.2.2 Attributes

The NearRTRICFunction IOC includes the attributes below and those attributes inherited through *ManagedFunction* IOC (defined in 3GPP TS 28.622 [3]).

Table 5.3.2.2-1: NearRTRICFunction attributes

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
pLMNIdentity	M	T	T	F	T
nearRTRICID	M	T	T	F	T
Attribute related to role					
applicationDNList	M	T	F	F	T

5.3.2.3 Attribute constraints

None

5.3.2.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.2.5 State diagram

None

5.3.3 EP_E2

5.3.3.1 Definition

The `EP_E2` IOC represents the management aspects of the E2 Termination (defined in O-RAN Near-RT RIC Architecture [6] clause 6.2.7.1).

5.3.3.2 Attributes

The `EP_E2` IOC includes the attributes below and those attributes inherited through *`EP_RP`* IOC (defined in 3GPP TS 28.622 [3]).

Table 5.3.3.2-1: EP_E2 attributes

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
tRICEventCreate	O	T	T	F	T
tRICEventDelete	O	T	T	F	T
tRICControl	O	T	T	F	T
tRICEventModify	O	T	T	F	T
tRICQuery	O	T	T	F	T

5.3.3.3 Attribute constraints

None

5.3.3.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.3.5 State diagram

None

5.3.4 EP_A1

5.3.4.1 Definition

The EP_A1 IOC represents the management aspects of the A1 Termination (defined in O-RAN Near-RT RIC Architecture [6] clause 6.2.7.2).

5.3.4.2 Attributes

The EP_A1 IOC includes the attributes inherited through *EP_RP* IOC (defined in 3GPP TS 28.622 [3]).

5.3.4.3 Attribute constraints

None

5.3.4.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.4.5 State diagram

None

5.3.5 NESPolicy

5.3.5.1 Definition

The NESPolicy IOC represents the policy for network energy saving (NES) for the NR Cell(s) managed by the O-DU.

When the NES Policy is created or updated by receiving a NES policy file, the NESPolicy MOI shall be created by MnS producer (O-DU) when it successfully receives NES policy file. The attribute policyId represents the unique identifier of the policy, corresponding to the policy file. The NES policy is applicable for the energy saving in the downlink.

An NES policy can be associated with none, one or more NR Cell(s) (i.e., NRCellDU) of the O-DU. When an NES policy is associated to none of NR Cell(s) (i.e., the DN of the NESPolicy MOI is not referenced in attribute configuredNesPolicyList of any NESPolicyRelation MOIs), the O-DU shall not consider the policy for evaluation.

The NESPolicy MOI corresponding to NES policy related to TRx Control (RF Channel Reconfiguration) based energy saving contains attribute policyType with value TRX_CONTROL. The attributes antennaMask and sleepMode provide the values from the NES policy file for the TRx control based energy saving.

The NESPolicy MOI corresponding to NES policy related to advanced sleep mode based energy saving contains attribute policyType with value ASM. The attribute sleepMode provides the value from the NES policy file for the advanced sleep mode based energy saving.

The MnS consumer shall be able to configure the time window(s) when this NES policy is applicable, using the attribute applicableTimeWindows of corresponding NESPolicy MOI. Multiple time windows can be configured for the policy using the attribute applicableTimeWindows. If the applicableTimeWindows is not present or if the startTime and endTime of any element of the list attribute applicableTimeWindows are configured with the same values, then O-DU shall not use the time condition for the policy, i.e., the policy is applicable without time condition. Setting overlapping time windows in the same NESPolicy MOI may cause undefined behavior.

5.3.5.2 Attributes

The NESPolicy IOC includes attributes inherited from *Top* IOC (defined in 3GPP TS 28.622 [3], clause 4.3.29) and the following attributes:

Table 5.3.5.2-1: NESPolicy attributes

Attribute name	S	isReadable	isWritable	isInvariant	isNotifyable
policyId	M	T	F	T	F
policyType	M	T	F	T	F
applicableTimeWindows	O	T	T	F	T
sleepMode	O	T	F	T	F
antennaMask	O	T	F	T	F

5.3.5.3 Attributes constraints

None

5.3.5.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.5.5 State diagram

None

5.3.6 NESPolicyRelation

5.3.6.1 Definition

The NESPolicyRelation IOC represents the NES policies configured for the NR Cell. NESPolicyRelation MOI with attribute configuredNesPolicyList with an empty list is invalid.

5.3.6.2 Attributes

The NESPolicyRelation IOC includes attributes inherited from *Top* IOC (defined in 3GPP TS 28.622 [3], clause 4.3.29) and the following attributes:

Table 5.3.6.2-1: NESPolicyRelation attributes

Attribute name	S	isReadable	isWritable	isInvariant	isNotifyable
Attribute related to role					
configuredNesPolicyList	M	T	T	F	T

5.3.6.3 Attributes constraints

None

5.3.6.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.6.5 State diagram

None

5.3.7 RRMPolicyRBAlloc

5.3.7.1 Definition

The `RRMPolicyRBAlloc` IOC provides the necessary attributes to set the resource segment defined as the combination of starting position of common resource block (CRB) allocation and number of physical resource blocks within the available frequency resource for each cell.

5.3.7.2 Attributes

The `RRMPolicyRBAlloc` IOC includes attributes inherited from *Top* IOC (defined in 3GPP TS 28.622 [3], clause 4.3.29) and contains the following attributes:

Table 5.3.7.2-1: RRMPolicyRBAlloc attributes

Attribute name	S	isReadable	isWritable	isInvariant	isNotifyable
direction	M	T	T	F	T
rRMPolicyMemberList	M	T	T	F	T
startCellRBAlloc	M	T	T	F	T
numberOfPRBs	O	T	T	F	T

5.3.7.3 Attribute constraints

None

5.3.7.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.7.5 State diagram

None

5.3.8 EP_D2C

5.3.8.1 Definition

This IOC represents the local end point of the control plane interface (D2-C) between two O-DUs. The transport network layer is based on IP transport with the SCTP on top of IP.

The application layer signalling protocol is referred to as D2CP (D2 Control Plane Protocol). See clause 7.1 of O-RAN.WG5.TS.D2GAP:D2 Interface [18].

NOTE 1: The existence of this MOI does not imply a successful D2 Interface establishment.

NOTE 2: This IOC is applicable only for D2 Interface for Intra CU inter DU carrier aggregation as defined in O-RAN.WG5.TS.D2GAP:D2 Interface [18].

5.3.8.2 Attributes

The EP_D2C IOC includes the attributes below and those attributes inherited through *EP_RP* IOC (defined in 3GPP TS 28.622 [3]).

Table 5.3.8.2-1: EP_D2C attributes

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
localAddress	O	T	T	T	T
localPortNumber	O	T	T	T	T
remoteAddress	O	T	T	T	T
remotePortNumber	O	T	T	T	T
tRemoteT1	O	T	T	T	T
tRemoteT2	O	T	T	T	T
tRemoteT3	O	T	T	T	T

5.3.8.3 Attribute constraints

None

5.3.8.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.8.5 State diagram

None

5.3.9 EP_D2U

5.3.9.1 Definition

This IOC represents the local end point of the user plane interface (D2-U) between two O-DUs. The transport network layer is based on IP transport with the UDP on top of IP.

The application layer signalling protocol is referred to as D2UP (D2 User Plane Protocol). See clause 7.2 of O-RAN.WG5.TS.D2GAP:D2 Interface [18].

NOTE: For each EP_D2U MOI, a corresponding EP_D2C MOI always exists associated with the same GNBDUNeighbour

5.3.9.2 Attributes

The EP_D2U IOC includes the attributes below and those attributes inherited through *EP_RP* IOC (defined in 3GPP TS 28.622 [3]).

Table 5.3.9.2-1: EP_D2U attributes

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
localAddress	O	T	T	T	T
localPortNumber	O	T	T	T	T
remoteAddress	O	T	F	F	T
remotePortNumber	O	T	F	F	T

5.3.9.3 Attribute constraints

None.

5.3.9.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.9.5 State diagram

None.

5.3.10 GNBDUNeighbour <<ProxyClass>>

5.3.10.1 Definition

This IOC represents an <<IOC>> `GNBDUFunction` or <<IOC>> `ExternalGNBDUFunction` as defined in 3GPP TS 28.541[12].

5.3.10.2 Attributes

See respective IOCs.

5.3.10.3 Attribute constraints

See respective IOCs.

5.3.10.4 Notifications

See respective IOCs.

5.3.10.5 State diagram

See respective IOCs.

5.3.11 D2Params

5.3.11.1 Definition

The `D2Params` IOC defines necessary attribute(s) that represents the properties of the O-DU supporting D2 interface.

5.3.11.2 Attributes

Table 5.3.11.2-1: D2Params attributes

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
t1	M	T	T	T	T
t2	M	T	T	T	T
t3	M	T	T	T	T
startCRNTI	O	T	T	F	T
endCRNTI	O	T	T	F	T
cellList	M	T	T	F	T
tClearTB	M	T	T	F	T

5.3.11.3 Attribute constraints

None

5.3.11.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.11.5 State diagram

None

5.3.12 ORU

5.3.12.1 Definition

The `ORU` IOC represents an O-RU (as defined in O-RAN Architecture Description [9], clause 5.3.6).

5.3.12.2 Attributes

The `ORU` IOC includes attributes inherited from *`Top`* IOC (defined in 3GPP TS 28.622 [3], clause 4.3.29) and the following attributes:

Table 5.3.12.2-1: ORU attributes

Attribute name	S	isReadable	isWritable	isInvariant	isNotifyable
`ruInstanceId`	M	T	T	T	T
Attribute related to role					
`oRUControllerList`	M	T	T	T	T

5.3.12.3 Attributes constraints

None

5.3.12.4 Notifications

The common notifications defined in clause 5.5 are valid for this class, without exceptions or additions.

5.3.12.5 State diagram

None

5.4 Attribute definitions

5.4.1 Attribute properties

The following table defines the properties of attributes that are specified in the present document.

Table 5.4.1-1: Attribute definitions

Attribute Name	Documentation and Allowed Values	Properties
`applicationVersion`	This attribute contains the application version numbers that shall consist of at least 3 fields, following a MAJOR.MINOR.PATCH pattern according to the Semantic Versioning Specification [15]. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
`applicationName`	This attribute contains the name of the application. allowedValues: N/A	type: String multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
`userLabel`	This attribute contains user defined label for the application allowedValues: N/A	type: String multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
`operationalState`	It indicates the operational state of the object instance. "It describes whether or not the resource is physically installed and working." This attribute is READ-ONLY. The meaning of these values is as defined in ITU-T Recommendation X.731 [8]. allowedValues: “ENABLED”, “DISABLED”	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
`usageState`	It indicates the usage state of the object instance. "It describes whether or not the resource is actively in use at a specific instant, and if so, whether or not it has spare capacity for additional users at that instant." This attribute is READ-ONLY. The meaning of these values is as defined in ITU-T Recommendation X.731 [8]. allowedValues: "IDLE", "ACTIVE", "BUSY"	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
`administrativeState`	It indicates the administrative state of the object instance. "It describes the permission to use or prohibition against using the resource, imposed through the management services." The meaning of these values is as defined in ITU-T Recommendation X.731 [8]. allowedValues: "LOCKED", "SHUTTINGDOWN", "UNLOCKED".	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
`tRICEventCreate`	Near-RT RIC attribute defined in O-RAN.WG3.TS.E2AP [5], clause 9.5 Specifies the maximum time for the RIC Subscription Request event creation procedure in the Near-RT RIC. allowedValues: [0..65535] ms	type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True

Attribute Name	Documentation and Allowed Values	Properties
tRICEventDelete	Near-RT RIC attribute defined in O-RAN.WG3.TS.E2AP [5], clause 9.5 Specifies the maximum time for the RIC Subscription Request event deletion procedure in the Near-RT RIC. allowedValues: [0..65535] ms	type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
tRICControl	Near-RT RIC attribute defined in O-RAN.WG3.TS.E2AP [5], clause 9.5 Specifies the maximum time for the RIC Control Request event request procedure in the Near-RT RIC. allowedValues: [0..65535] ms	type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
tRICEventModify	Near-RT RIC attribute defined in O-RAN.WG3.TS.E2AP [5], Clause 9.5 Specifies the maximum time for the RIC Subscription Modification procedure in the Near-RT RIC. allowedValues: [0..65535] ms	type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
tRICQuery	Near-RT RIC attribute defined in O-RAN.WG3.TS.E2AP [5], Clause 9.5 Specifies the maximum time for the RIC Query procedure in the Near-RT RIC. allowedValues: [0..65535] ms	type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
nearRTRICID	Defined in O-RAN.WG3.TS.E2AP [5], clause 9.2.4 allowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
pLMNIdentity	Defined in 3GPP TS 28.658 [7], clause 4.3.26 allowedValues: N/A	type: PLMNId multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
policyId	It defines the unique identifier of the policy, corresponding to the policy file. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
policyType	It indicates the type of energy saving corresponding to NES policy. allowedValues: "TRX_CONTROL": the policy is for TRx Control (RF Channel Reconfiguration) based energy saving. "ASM": the policy is for advanced sleep mode based energy saving.	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False

Attribute Name	Documentation and Allowed Values	Properties
applicableTimeWindows	It defines the list of time windows at which the NES policy is applicable. allowedValues: N/A	type: TimeWindow multiplicity: 1..* isOrdered: False isUnique: True defaultValue: None isNullable: False
sleepMode	It indicates the sleep mode (defined in O-RAN.WG4.TS.CUS.0 specification [13], clause 7.5.3.52) corresponding to NES policy. Allowed values are the sleep modes supported by the O-RU (defined in O-RAN.WG4.TS.CUS.0 specification [13], clause 16.1). allowedValues: 0, 1, 2, 3	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
antennaMask	It indicates or defines the antenna mask (defined in O-RAN.WG4.TS.CUS.0 specification [13], clause 7.5.3.54) corresponding to NES policy. allowedValues: N/A	type: BitString multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
configuredNesPolicyList	It holds an ordered list of DNs of the NES policies that are configured for a given NRCellDU. The evaluation of the NES policy is performed by the O-DU in increasing order of their precedence in the list. The order represents the policy priorities, i.e., the policy whose DN appears first has higher priority than the policies that follow. allowedValues: DNs of the NESPolicy MOI(s) that are name contained in the parent GNBDUFunction.	type: DN multiplicity: 1..* isOrdered: True isUnique: True defaultValue: None isNullable: False
startCellRBAlloc	Specifies the starting position of common resource block (CRB) allocation within the available frequency resource for each cell. This value is the offset in common resource blocks, defined in 3GPP TS 38.211[14] clause 4.4.4.3, to common resource block 0 for the applied subcarrier spacing for a cell. allowedValues: 0 to N_grid_size - 1, where N_grid_size equals the number of common resource blocks for the BS channel bandwidth based on the subcarrier spacing applied to the cell.	type: integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: 0 isNullable: False
direction	The direction of interest for a startCellRBAlloc. allowedValues: BIDIRECTION, UL, DL	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: BIDIRECTION isNullable: False
rRMPolicyMemberList	It represents the list of RRMPolicyMember (s) that the managed object is supporting, defined in 3GPP TS 28.541[12] clause 4.3.42. A RRMPolicyMember <<dataType>> includes the PLMNId <<dataType>> and S-NSSAI <<dataType>>. allowedValues: N/A	type: RRMPolicyMember multiplicity: 1..* isOrdered: False isUnique: True defaultValue: None isNullable: False

Attribute Name	Documentation and Allowed Values	Properties
numberOfPRBs	Number of physical resource blocks, defined in 3GPP TS 38.211 [14] clause 4.4.4.4, for a RRMPolicyRBAlloc. allowedValues: 1 to N_grid_size – startCellRBAlloc. See startCellRBAlloc for definition of N_grid_size. This applies only to single numerology use case.	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
hostDN	This attribute contains the DN of the hosting entity (e.g., NearRTRICFunction). allowedValues: N/A	type: DN multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
applicationDNList	This attribute contains the DNs of the hosted applications (MOIs of the concrete IOCs inheriting from *ManagedApplication*). allowedValues: N/A	type: DN multiplicity: 0..* isOrdered: False isUnique: True defaultValue: None isNullable: True
localAddress	This parameter specifies the localAddress used for initialization of the underlying transport. The AddressWithVlan <<dataType>> is defined in clause 4.3.64 of 3GPP TS 28.541 [12] AllowedValues: N/A	type: AddressWithVlan multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
remoteAddress	Remote address including IP address used for initialization of the underlying transport. IP address can be an IPv4 address (See IETF RFC 791 [22]) or an IPv6 address (See IETF RFC 2373 [23]). AllowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: 0.0.0.0 for IPv4 or :: for IPv6 isNullable: False
localPortNumber	It defines the port number used by the local end entity of an end point. AllowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
remotePortNumber	It defines the port number used by the far end entity of an end point. AllowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: 0 isNullable: False
tRemoteT1	This attribute pertains to the timer value T1 applicable at the far end O-DU as defined in O-RAN.WG5.TS-D2-O&MRequirements [19], clause 4.1.1.1. It is an unsigned integer identifying a period of time in units of microseconds. allowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False

tRemoteT2	This attribute pertains to the timer value T2 applicable at the far end O-DU as defined in O-RAN.WG5.TS-D2-O&MRequirements [19], clause 4.1.1.2. It is an unsigned integer identifying a period of time in units of microseconds. allowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
tRemoteT3	This attribute pertains to the timer value T3 applicable at the far end O-DU as defined in O-RAN.WG5.TS-D2-O&MRequirements [19], clause 4.1.1.3. It is an unsigned integer identifying a period of time in units of microseconds. allowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
t1	Minimum time for SCell O-DU to process and transmit data to a UE once data is received from PCell O-DU on D2 interface. See clause 4.1.1.1 of O-RAN.WG5.TS-D2-O&MRequirements [19], It is an unsigned integer identifying the period of time in units of microseconds. allowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
t2	Minimum time for PCell O-DU to process and respond to DL DATA REQUIRED message, and subsequent transmission of data in the OTA slot. See clause 4.1.1.2 of O-RAN.WG5.TS-D2-O&MRequirements [19]. It is an unsigned integer identifying the period of time in units of microseconds. allowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
t3	Maximum time within which PCell O-DU sends a DL SCHEDULING REQUEST message requesting scheduling for a given OTA slot. See clause 4.1.1.3 of O-RAN.WG5.TS-D2-O&MRequirements [19]. It is an unsigned integer identifying the period of time in units of microseconds. allowedValues: N/A	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
startCRNTI	This attribute defines the start C-RNTI assigned to this O-DU. See clause 4.1.1 (Start CRNTI) of O-RAN.WG5.TS-D2-O&MRequirements [19]. allowedValues: 1..65522	type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
endCRNTI	This attribute defines the end C-RNTI assigned to this O-DU. See clause 4.1.2 (CRNTI Range) of O-RAN.WG5.TS-D2-O&MRequirements [19] allowedValues: 1..65522	type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True

`cellList`	This is a list of cells supported by an O-DU that can also support inter O-DU CA via D2 interface. It specifies the cell local identifier for the cell that supports D2 interface. This attribute along with the gNB Identifier (using `gNBId` of the parent `GNBDUFunction`), identifies a NR cell within a PLMN. This is the NR Cell Identity (NCI). See subclause 8.2 of 3GPP TS 38.300 [21]. allowedValues: 0..16383	type: Integer multiplicity: 1..* isOrdered: False isUnique: True defaultValue: None isNullable: False
`tClearTB`	This parameter specifies the maximum duration for which the SCell O-DU retains the transport Block (TB) for retransmission following a HARQ failure. See subclause 5.4.3.31 of O-RAN.WG5.TS.D2AP [20] allowedValues: 1ms, 2ms, 5ms, 10ms, 20ms, 60ms	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
`ruInstanceId`	Identifies the O-RU as defined in Open FH M-Plane Specification [24]. The format is as specified in Open FH M-Plane Interface specification [24]: urn:o-ran:operations:1.0:operational-info:declarations:ru-instance-id allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
`oRUControllerList`	This attribute contains the DNs of the `gNBDUFunction` instances with an Open Fronthaul interface to the O-RU. allowedValues: N/A	type: DN multiplicity: 1..* isOrdered: False isUnique: True defaultValue: None isNullable: False

5.5 Common notifications

5.5.1 Alarm notifications

This clause presents a list of notifications, defined in 3GPP TS 28.111 [16] and referenced in O-RAN O1 Interface Specification [11], that an MnS consumer may receive. The notification header attribute `objectClass/objectInstance`, defined in 3GPP TS 28.111 [16], shall capture the DN of an instance of a class defined in the present document.

Table 5.5.1-1: Alarm notifications

Name	S	Notes
`notifyNewAlarm`	M	
`notifyClearedAlarm`	M	
`notifyAckStateChanged`	O	
`notifyAlarmListRebuilt`	M	
`notifyChangedAlarm`	O	
`notifyChangedAlarmGeneral`	O	

5.5.2 Configuration notifications

This clause presents a list of notifications, defined in 3GPP TS 28.532 [10] and referenced in O-RAN O1 Interface Specification [11], that an MnS consumer may receive. The notification header attribute `objectClass/objectInstance`, defined in 3GPP TS 28.532 [10], shall capture the DN of an instance of a class defined in the present document.

Table 5.5.2-1: Configuration notifications

Name	S	Notes
`notifyMOIChanges`	O	
`notifyEvent`	O	

Annex (informative): Change History

Date	Revision	Description
2024.07.26	01.00	First Release Inclusion of CRs: • O1NRM Requirements • Common Nofications • O1 NRM Models addtions including RIC • O1 NRM Scope addition • O1 NRM Terms addition • O1 NRM Short name change • O1 NRM Spec O-DU IM for NES • O1 Interface Enhancement for Inter-cell Interference Suppression Control • MA and RIC Model update • Bi-Directional EP_E2 Model • O1 NRM addition WG3 Params • O1 NRM addition of EP_A1 Editorial fixes and alignments.
2024.12.06	02.00	Inclusion of CRs: • Add references to the Clause 5.5 “Common notifications” for applicable IOCs • Change the cardinality to allow a cell to exist without an instance of RRMPolicyRBAlloc • Various updates to references as needed • Various updates to fonts etc • Delete all import related notes • Adding requirements to follow 3GPP TS 32.160 clause 5.1 • Adding new attribte “numberOfPRBs” with its definition • Change 3GPP references boiler plate text to Rel 18.
2025.03.10	03.00	Inclusion of CRs: • Requirements update, Improve mapping rule from Stage 2 to Stage 3
2025.07.21	04.00	Inclusion of CRs • Corrections and editorial updates Further editorial modifications
2026.03.16	05.00	Inclusion of CRs: • O-DU D2 Configuration parameters • EPs for D2Interface • ORU IOC Introduction Minor editorial updates

